

Retail Demand Forecasting

Project Documentation

Table of Contents

1. Project Overview
2. Objectives
3. Dataset Information
4. Technologies Used
5. System Requirements
6. Project Workflow
7. Data Preprocessing
8. Exploratory Data Analysis
9. Forecasting Model
10. Streamlit Dashboard
11. Project Output
12. How to Run the Project
13. Conclusion
14. Future Scope

1. Project Overview

Retail demand forecasting predicts future product demand using historical sales data. This project uses the Facebook Prophet model and a Streamlit dashboard to forecast retail sales and support inventory planning.

2. Objectives

- Analyze historical retail sales data.
- Clean and preprocess the dataset.
- Perform exploratory data analysis.
- Forecast sales for the next 30 days.
- Develop an interactive Streamlit dashboard.
- Support inventory planning.

3. Dataset Information

Dataset: M5 Forecasting Dataset

Domain: Retail Sales Forecasting

Forecast Horizon: 30 Days

The dataset contains historical retail sales, calendar information, and selling prices.

4. Technologies Used

Python, Pandas, NumPy, Matplotlib, Facebook Prophet, Streamlit, and Jupyter Notebook.

5. System Requirements

Hardware: Minimum 8 GB RAM, Intel i3 or above.

Software: Windows OS, Python 3.x, Jupyter Notebook, Streamlit.

6. Project Workflow

Data Collection → Data Cleaning → EDA → Data Preparation → Prophet Model Training → Forecast Generation → Dashboard Development.

7. Data Preprocessing

Handled missing values, formatted date columns, prepared the data for forecasting, and validated the input dataset.

8. Exploratory Data Analysis

Analyzed top-selling products, daily sales, selling prices, product categories, event distribution, weekday distribution, and monthly trends.

9. Forecasting Model

Facebook Prophet was used to generate a 30-day sales forecast from historical retail sales data.

10. Streamlit Dashboard

The dashboard includes forecast data, a forecast chart, sales summary, inventory recommendation, dataset information, and model information.

11. Project Output

The project generates an interactive dashboard that helps users visualize future demand and supports inventory planning.

12. How to Run the Project

Install required libraries, execute the Jupyter Notebook

Run 'streamlit run app.py' to launch the dashboard.

13. Conclusion

The project demonstrates an effective application of time-series forecasting for retail demand prediction.

14. Future Scope

Future improvements include cloud deployment, real-time forecasting, and comparison with additional forecasting models.